Five number summary

1 For each of the the following sets of data:

i Sort the data in ascending order.

ii Find the maximum value.

iii Find the minimum value.

iv Find the median value.

v Find Q_1 for this data set.

vi Find Q_3 for this data set.

a The data set shows finishing times (in minutes) of the competitors in a 1500-meter swimming race:

$$24.41,\ 22.95,\ 21.88,\ 24.19,\ 16.12,\ 25.64,\ 16.83,\ 23.62,\ 24.52,\ 23.74,\ 19.44$$

b The data set shows number of points scored by a basketball team in each game of their previous season:

$$75,\ 53,\ 84,\ 66,\ 89,\ 55,\ 63,\ 70,\ 92,\ 51,\ 90,\ 55,\ 81,\ 87,\ 68$$

c The data set shows marks in an end-of-year exam for a class of students:

$$59,\ 53,\ 75,\ 80,\ 82,\ 96,\ 81,\ 79,\ 64,\ 58,\ 77,\ 62,\ 62,\ 86$$

2 There is a test to measure the Emotional Quotient (EQ) of an individual. Here are the EQ results for 21 people, listed in ascending order:

$$92,\ 94,\ 100,\ 103,\ 103,\ 105,\ 105,\ 109,\ 110,\ 113,\ 114,$$
$$114,\ 116,\ 118,\ 118,\ 119,\ 120,\ 125,\ 125,\ 126,\ 130$$

a Determine the median EQ score.

b Determine Q_1 for this data set.

c Determine Q_3 for this data set.

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3 An advertising agency recorded the number of viewers within various age ranges of its newest television advertisement when it went to air. The results are shown in the table:

Age Interval	Frequency	Mean Age
$16-20$	350	18
$21-25$	150	23
$26-30$	200	28
$31-35$	300	33
$36-40$	300	38
$41-45$	300	43
$46-50$	400	48
$51-55$	300	53
$56-60$	200	58
$61-65$	50	63

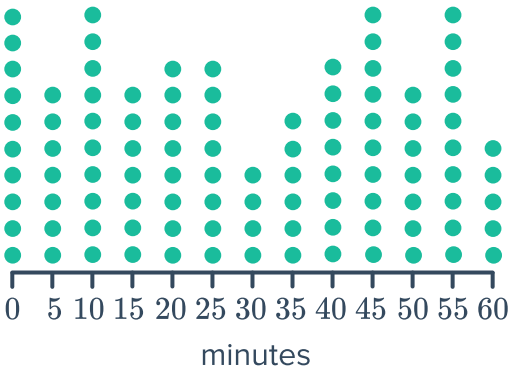
a Using the mean age of each age interval, find the five number summary.

b According to the five-number summary, approximately what percentage of viewers of the ad were aged between 28 and 48?

c The advertising agency was targeting viewers aged between 18 and 28. They would deem their ad as successful if at least 60% of viewers of the ad were in this age range.

According to the five number summary, were they successful in reaching their target viewers?

4 The airline Flo Air decided to keep track of flight delay times (the number of minutes after the scheduled time when the plane takes off) over a week. The 100 results are shown in the dot plot:



a Determine the median delay time of the flights, in minutes.

b Determine the upper quartile.

c Determine the lower quartile.

d Determine the interquartile range.

e If a flight is delayed for 10 minutes or more, the airline incurs a fee. According to the given dot plot, for what percentage of flights did the airline incur a fee?

f A rival airline, Fly Air, had a mean delay time during the same week of 45 minutes. What percentage of Flo Air's flights had delay times that were longer than Fly Air's mean delay time?

5 If there are 78 scores in a set of data, in which position will the lower quartile lie?

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6 Consider the scores below. Between which values do the middle 50% of scores lie?

$$10, 11, 12, 13, 15, 17, 19, 20$$

7 In competition, a diver must complete 8 rounds of dives. Her scores for the first 7 rounds are:

$$7.3, 7.4, 7.7, 8.4, 8.7, 8.9, 9.4$$

Determine her score in the 8th round if the upper quartile of all of her 8 scores is 8.85.

8 To gain a place in the main race of a car rally, teams must compete in a qualifying round. The median time in the qualifying round determines the cut off time to make it through to the main race. Below are some results from the qualifying round:

- 75% of teams finished in 159 minutes or less.
- 25% of teams finished in 132 minutes or less.
- 25% of teams finished between with a time between 132 and 142 minutes.

a Find the median time for the qualifying round.

b Hence, state the cut off time required in the qualifying round to make it through to the main race.

c Determine the interquartile range in the qualifying round.

d In the qualifying round, the ground was wet, while in the main race, the ground was dry. To make the times more comparable, the finishing time of each team from the qualifying round is reduced by 5 minutes.

Find the new median time from the qualifying round.

9 The table shows the luggage weight, in kilograms, of 30 passengers.

Weight	Frequency
16	5
17	5
18	2
19	4
20	6
21	4
22	4

a What is the mean check in weight? Round your answer to two decimal places.

b Determine the median, Q_1, and Q_3.

c In which quartile does the mean lie?

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Box-and-whisker plots

10 For the box-and-whisker plot shown below, find each of the following:

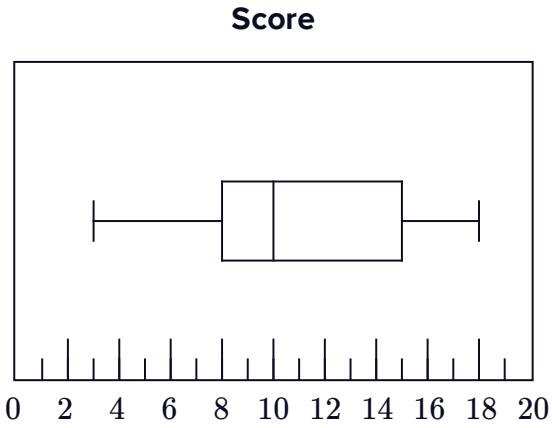
a Lowest score

b Highest score

c Range

d Median

e Interquartile range



11 Construct a box-and-whisker plot for each five number summary:

a

- Median $= 47$
- Lower Quartile $= 33$
- Upper Quartile $= 61$
- Lowest score $= 16$
- Highest score $= 71$

b

- Median $= 36$
- Lower Quartile $= 28$
- Upper Quartile $= 42$
- Lowest score $= 20$
- Highest score $= 52$

c

- Median $= 35$
- Lower Quartile $= 25$
- Upper Quartile $= 60$
- Lowest score $= 5$
- Highest score $= 75$

12 A geography teacher has marked a set of tests. She wants to represent the results in a box-and-whisker plot. She has already sorted her data and created the table shown. Create a box-and-whisker plot to match the data in the table:

Minimum	8
Lower Quartile	10
Median	16
Upper Quartile	24
Maximum	28

13 Consider the following set of data:

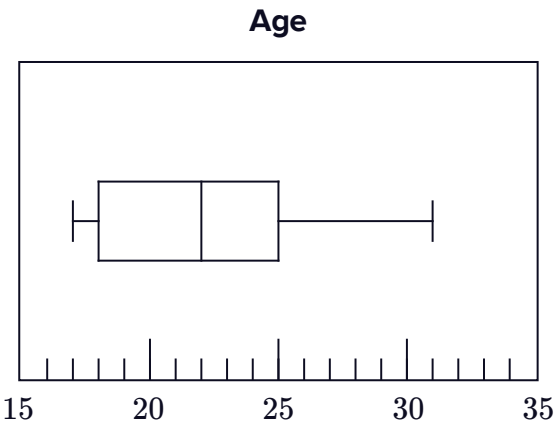
$$48, 16, 36, 32, 36, 36, 4, 16, 8$$

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a State the five number summary.

b Create a box-and-whisker plot to represent the data.

14 The box-and-whisker plot shows the age at which a group of people got their driving licences:



a What is the oldest age?

b What is the youngest age?

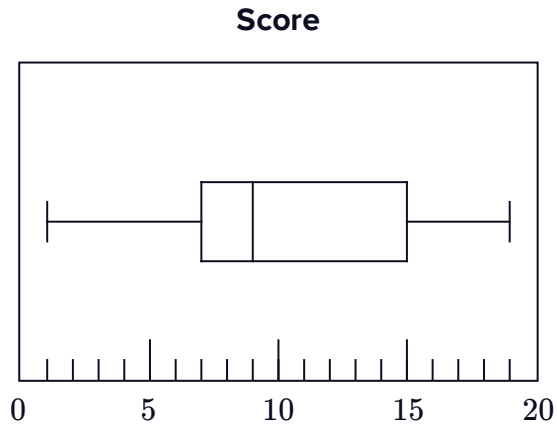
c What percentage of people were aged from 18 to 22?

d The middle 50% of responders were within how many years of one another?

e In which quartile are the ages least spread out?

f The bottom 50% of responders were within how many years of one another?

15 Consider the box-and-whisker plot shown:



a Determine what percentage of scores lie between each of the following:

i 7 and 15 inclusive

ii 1 and 7 inclusive

iii 19 and 9 inclusive

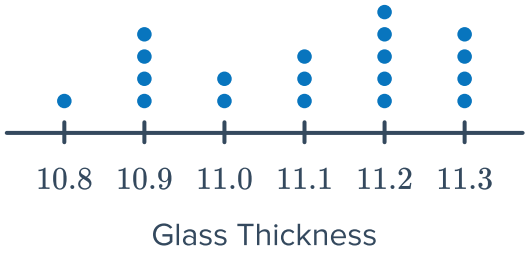
iv 7 and 19 inclusive

v 1 and 15 inclusive

b In which quartile is the data the least spread out?

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16 The glass windows for an airplane are cut to a certain thickness, but machine production means there is some variation. The thickness of each pane of glass produced is measured (in millimetres), and the dot plot shows the results:



a Find the median thickness, to two decimal places.

b Find the interquartile range.

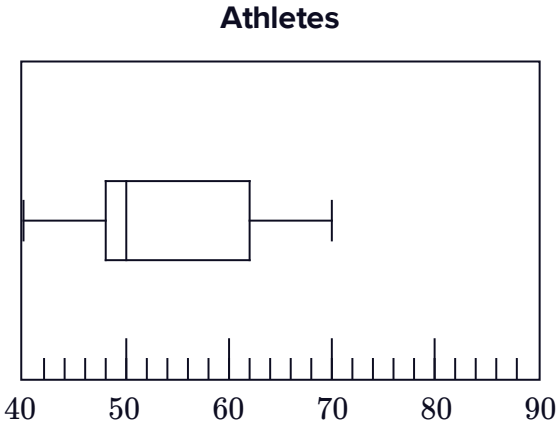
c Construct a box-and-whisker plot to represent the data.

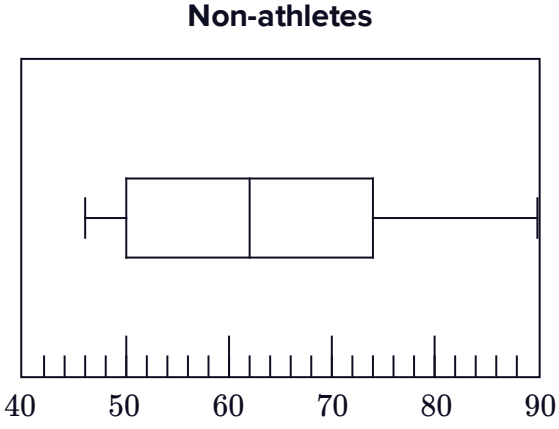
d What percentage of thicknesses were between 10.8 mm and 11.2 mm inclusive? Round your answer to two decimal places if necessary.

e According to the box-and-whisker plot, in which quartile are the results the most spread out?

f Which statistics cannot be found from a box-and-whisker plot?

17 Two groups of people, athletes and non-athletes, had their resting heart rate measured. The results are displayed in the given pair of box-and-whisker plots.





a What is the median heart rate of athletes?

b What is the median heart rate of the non-athletes?

c Using this measure, which group has the lower heart rates?

d What is the interquartile range of the athletes' heart rates?

e What is the interquartile range of the non-athletes' heart rates?

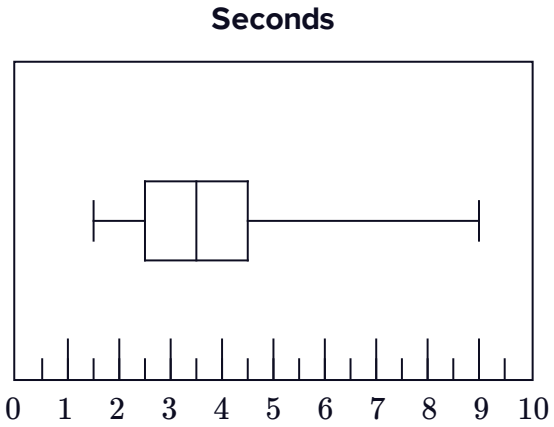
f Using this measure, which group has more consistent heart rate measures?

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18 In training, a fighter pilot measures the number of seconds he blacks out over a number of flights. He constructs the box-and-whisker plot shown:

As long as the pilot is not unconscious for more than 7 seconds, he will be safe to fly. The pilot concludes that he is safe to fly all the time.

Explain why his conclusion is incorrect.



Box-and-whisker plots with technology

19 For each of the following data sets:

i Use your CAS calculator to create a box-and-whisker plot.

ii Using the box-and-whisker plot or otherwise, find the five number summary for the data set.

a $25, 17, 16, 12, 26, 6, 32, 24, 47$

b $16.7, 8.5, 13.4, 17.5, 2, 9.3, 12.5, 7.9, 10.5, 14$

c $2, 10, 1, 20, 16, 18, 15, 4, 13$

20 Consider the following set of numbers: $4, 2, 6, 8, 11, 20, 15, 10, 11$

a Use your CAS calculator to create a box-and-whisker plot that represents the data set.

b Determine whether 11 represents the mean, median, mode, or an outlier.

c Using the box-and-whisker plot or otherwise, find the five number summary for the data set.

21 Consider the following set of numbers: $11, 4, 13, 7, 6, 8, 12, 45, 3, 10, 17$

a Use your CAS calculator to create a box-and-whisker plot that represents the data set.

b Determine whether 45 represents the mean, median, mode, or an outlier.

c Using the box-and-whisker plot or otherwise, find the five number summary for the data set.

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22 You conduct a straw poll with some of the students in your class, asking the question "How many times have you accessed social media so far today?".

The responses are summarised in the following list: $5, 4, 8, 4, 7, 9, 9, 1, 4, 5, 7$

a Use your CAS calculator to create a box-and-whisker plot for this data set.

b Determine whether 4 represents the mean, median, mode, or an outlier.

c Using the box-and-whisker plot or otherwise, find the five number summary for the data set.

23 You conduct a survey with a group of friends, asking the question "On average, how many hours per day do you spend using technology for entertainment purposes?".

The responses were given to the nearest half hour: $12.5, 12, 4.5, 13, 6.5, 9.5, 3, 9.5, 9, 4.5, 5.5$

a Use your CAS calculator to create a box-and-whisker plot for this data set.

b Does this data set contain any outliers?

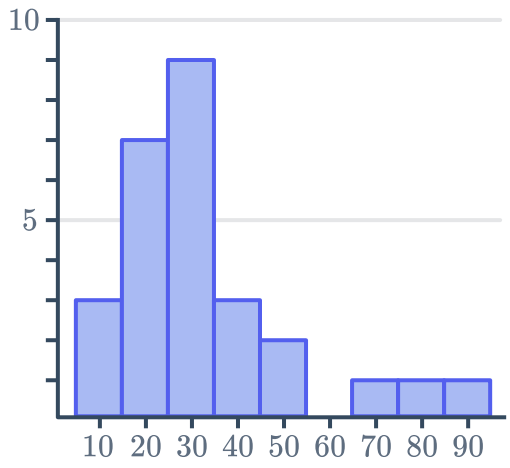
c Using the box-and-whisker plot or otherwise, find the five number summary for the data set.

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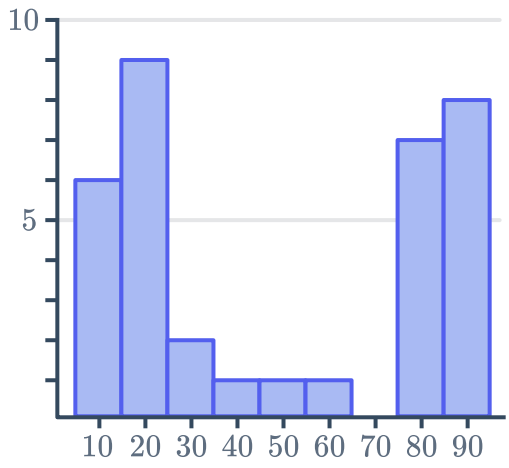
Box-and-whisker plots and the shape of data

24 Construct a box-and-whisker plot for the following histograms:

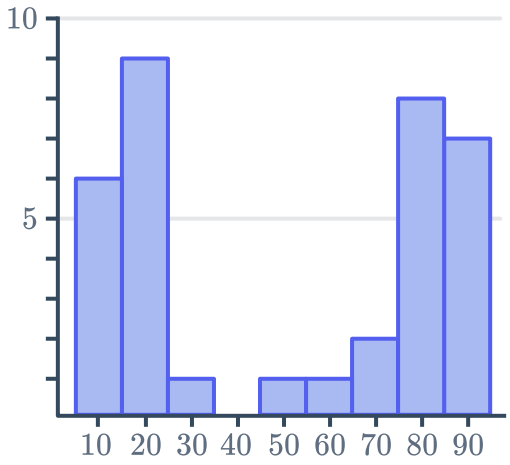
a



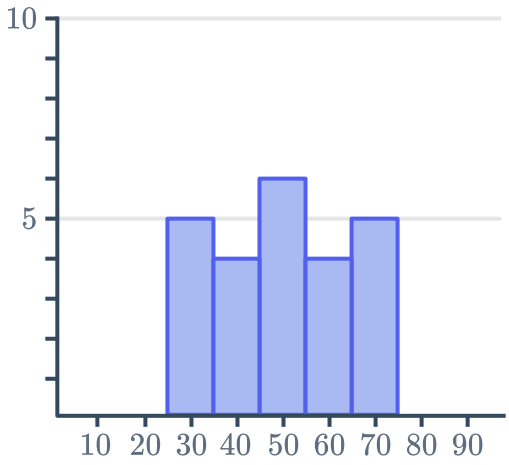
b



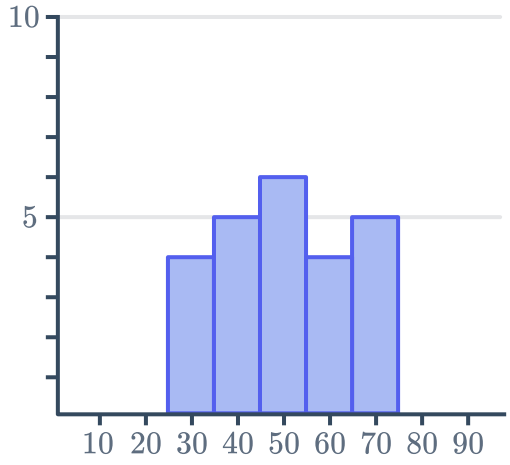
c



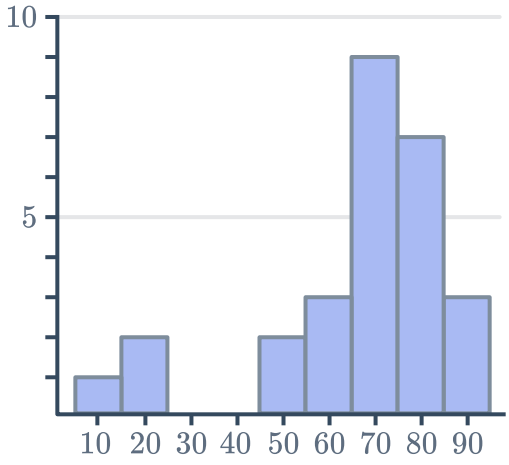
d



e



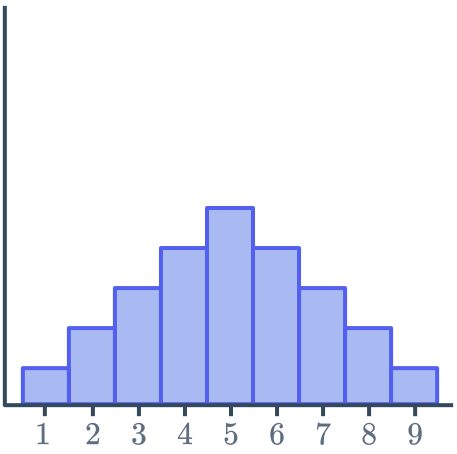
f



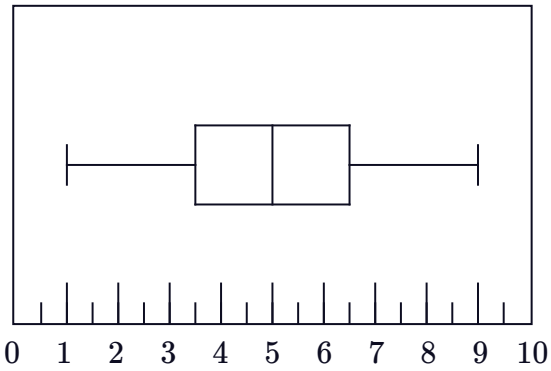
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25 Match the histograms on the left to the corresponding box-and-whisker plots on the right:

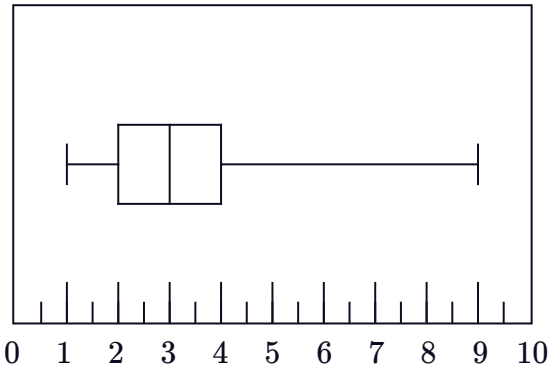
Histogram A



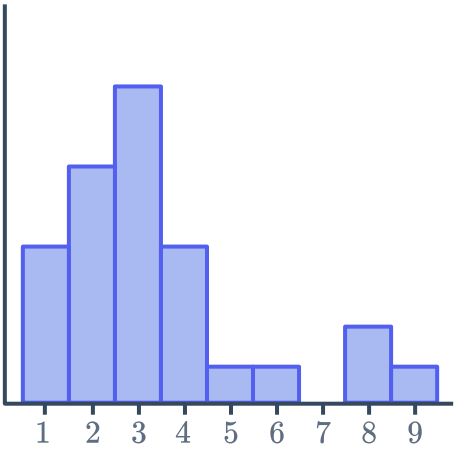
Box Plot 1



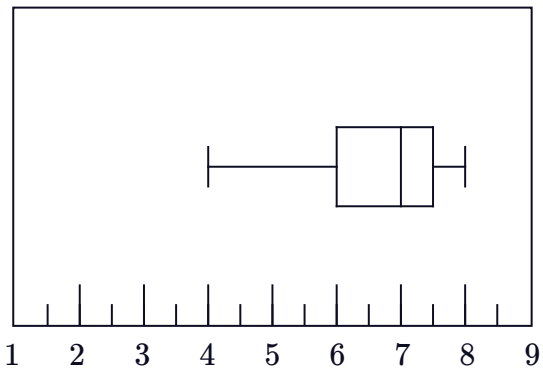
Box Plot 2



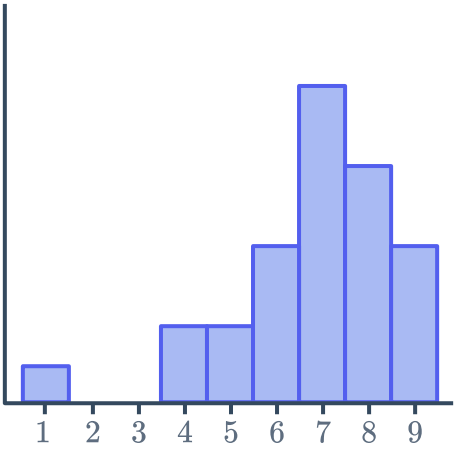
Histogram B



Box Plot 3

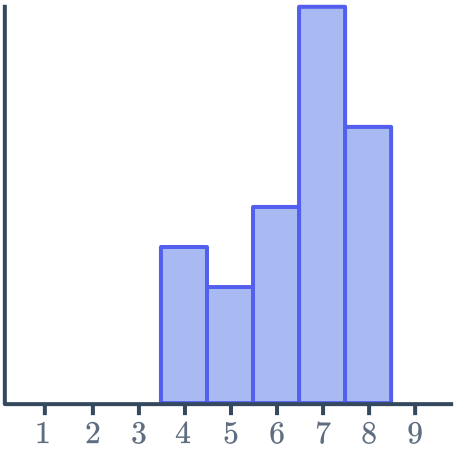


Histogram C

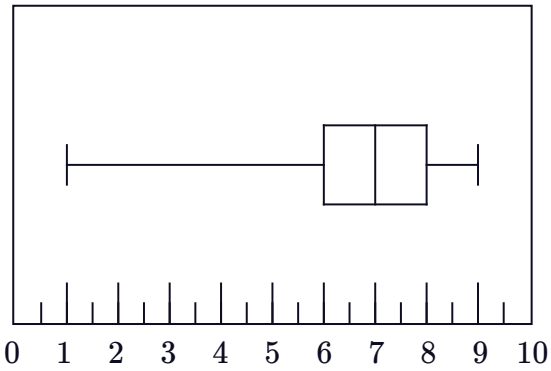


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Histogram D

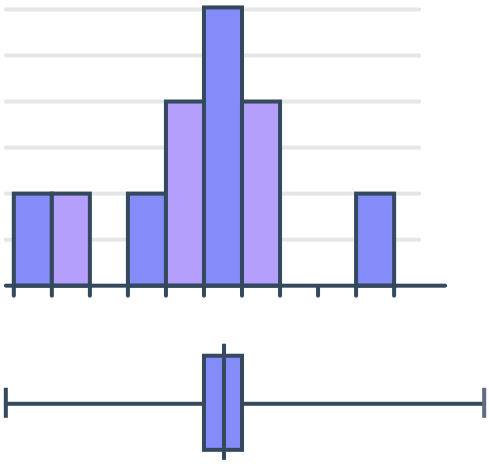


Box Plot 4

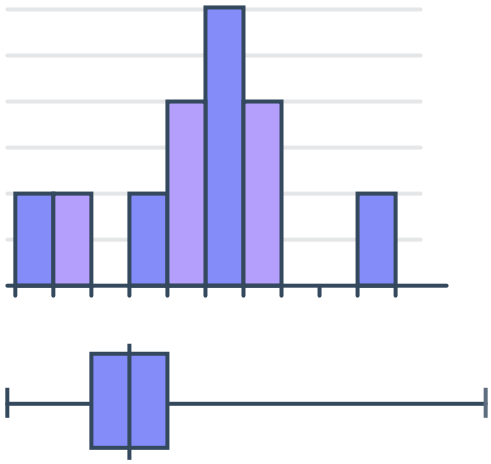


26 State whether the following pairs of histograms and box-and-whisker plots match with respect to their shape:

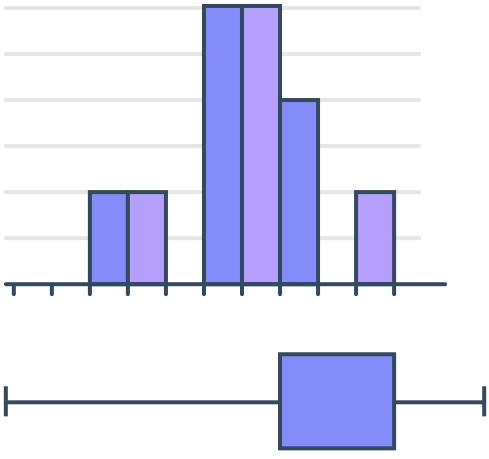
a



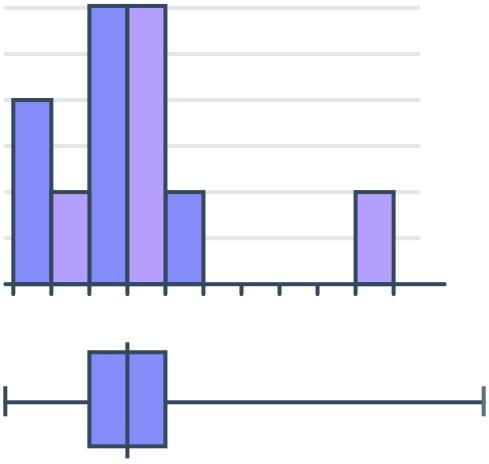
b



c

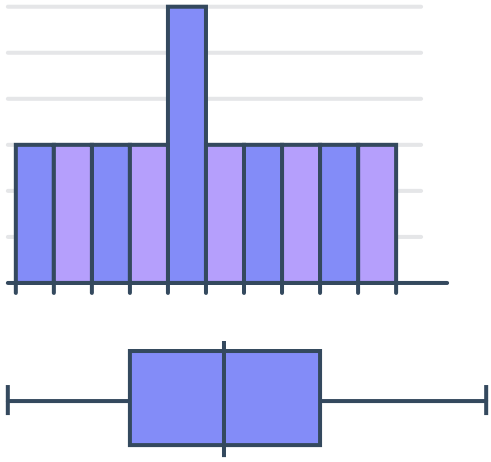


d

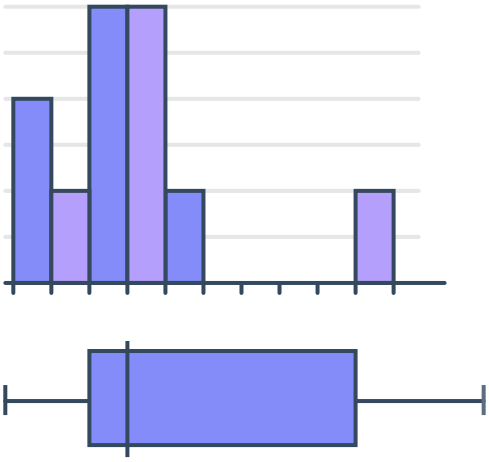


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e

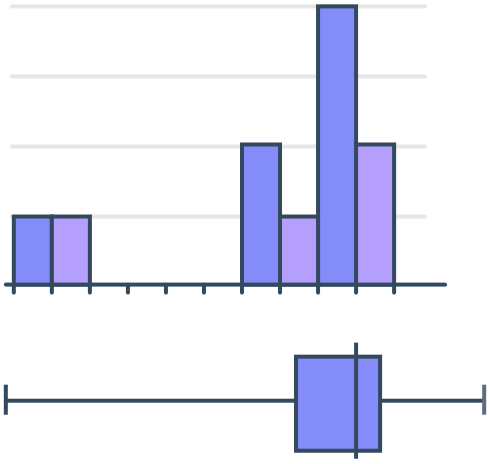


f

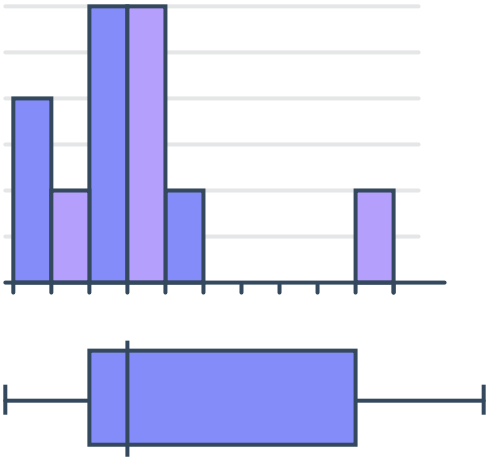


27 Explain why the following pairs of histograms and box-and-whisker plots do not match:

a



b



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